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Target: Parallelism TS 2

FINDING THE RIGHT SET OF TRAITS FOR `SIMD<T>`

ABSTRACT

This paper makes the set of traits for `simd<T>` more complete.

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1

INTRODUCTION

[N4744] defines the trait `simd_abi::deduce<T, N>`, allowing users to find an “implementation-recommended” ABI tag for a given `value_type` and number of elements. Shen [P0820R1] discusses a use for considering involved ABI tags in the “recommendation”. SG1 polled in Albuquerque about

Poll: `abi_for_size_t` (SF) vs. *implementation-defined* (SA)

SF	F	N	A	SA
1	7	7	0	0

The poll result implies that SG1 prefers users to be able to spell out the ABI tags that are determined as return types.

2

MOTIVATION

As Shen [P0820R1] shows, there is a use case for deducing an ABI tag type from a `value_type`, a width, and additionally zero or more “input” ABI tags. The latter tells the deduction logic what ABI tags are used in the input types to produce an object of the requested `value_type` and width. This enables an implementation design choice of staying within a certain SIMD register subset.

From the user’s perspective, the ABI tag deduction is most often necessary in the following two cases:

- Given a certain `simd` type, what is the best `simd` type for a different `value_type` (e.g. mixed precision calculations).
- Given a certain `simd` type, what is the best `simd` type for a different width (e.g. split, concat, shuffle).

Therefore, I propose to

1. extend `simd_abi::deduce` to consider input ABI tags in its decision,
2. introduce a new trait `rebind_simd<U, V>`, which deduces a `simd<U, Abi>` instantiation from a given `simd` type `V` and requested `value_type` `U`, and
3. introduce a new trait `resize_simd<N, V>`, which deduces a `simd<T, Abi>` instantiation from a given `simd` type `V` with `value_type` `T` and requested width `N`.

3

PROPOSED WORDING

Apply the following change to the Parallelism TS 2 [N4744]:

modify [parallel.simd.synopsis]

```
template <class T, size_t N> struct deduce { using type = see below; };
template <class T, size_t N> using deduce_t = typename deduce<T, N>::type;
template <class T, size_t N, class... Abis> struct deduce { using type = see below; };
template <class T, size_t N, class... Abis> using deduce_t = typename deduce<T, N, Abis...>::type;
```

add to [parallel.simd.synopsis]

```
inline constexpr size_t memory_alignment_v = memory_alignment<T, U>::value;

template <class T, class V> struct rebind_simd { using type = see below; };
template <class T, class V> using rebind_simd_t = typename rebind_simd<T, V>::type;
template <int N, class V> struct resize_simd { using type = see below; };
template <int N, class V> using resize_simd_t = typename resize_simd<N, V>::type;
```

modify [parallel.simd.abi]

```
template <class T, size_t N> struct deduce { using type = see below; };
template <class T, size_t N, class... Abis> struct deduce { using type = see below; };

```

- 12 The member type shall be present if and only if
- T is a vectorizable type, and
 - `simd_abi::fixed_size<N>` is supported (see 9.2.1), and
 - every type in the Abis pack is an ABI tag.
- 13 Where present, the member typedef type shall name an ABI tag type that satisfies
- `simd_size_v<T, type> == N`, and
 - `simd<T, type>` is default constructible (see 9.3.1),

If N is 1, the member typedef type is `simd_abi::scalar`. Otherwise, if there are multiple ABI tag types that satisfy the constraints, the member typedef type is implementation-defined. [*Note:* It is expected that extended ABI tags can produce better optimizations and thus are preferred over `simd_abi::fixed_size<N>`. Implementations can base the choice on Abis, but can also ignore the Abis arguments. — *end note*]

add at the end of [parallel.simd.traits]

```
template <class T, class V> struct rebind_simd { using type = see below; };

```

- 15 The member type shall be present if and only if
- V is either `simd<U, Abi0>` or `simd_mask<U, Abi0>`, where U and Abi0 are deduced from V, and

- T is a vectorizable type, and

- simd_abi::deduce<T, simd_size_v<U, Abi0>, Abi0> has a member type type.

16 Let Abi1 identify the type deduce_t<T, simd_size_v<U, Abi0>, Abi0>. Where present, the member typedef type shall name simd<T, Abi1> if V is simd<U, Abi0> or simd_mask<T, Abi1> if V is simd_mask<U, Abi0>.

template <int N, class V> struct resize_simd { using type = see below; };

17 The member type shall be present if and only if

- V is either simd<T, Abi0> or simd_mask<T, Abi0>, where T and Abi0 are deduced from V, and
- simd_abi::deduce<T, N, Abi0> has a member type type.

18 Let Abi1 identify the type deduce_t<T, N, Abi0>. Where present, the member typedef type shall name simd<T, Abi1> if V is simd<T, Abi0> or simd_mask<T, Abi1> if V is simd_mask<T, Abi0>.

4

CHANGELOG

4.1

CHANGES FROM R0

Previous revision: [P0964R0].

- Adjusted to changes between [P0214R8] and [N4744].
- Make `resize_simd` a non-optional part of the requested changes (after SG1 discussion).
- Update motivation after resolving different naming preferences with Tim.

5

STRAW POLLS

5.1

SG1 AT JACKSONVILLE 2018

Poll: Proceed to LEWG?

→ unanimous consent

A

BIBLIOGRAPHY

- [N4744] Jared Hoberock, ed. *Technical Specification for C++ Extensions for Parallelism Version 2*. ISO/IEC JTC1/SC22/WG21, 2018. URL: <https://wg21.link/n4744>.

- [P0214R8] Matthias Kretz. *P0214R8: Data-Parallel Vector Types & Operations*. ISO/IEC C++ Standards Committee Paper. 2018. URL: <https://wg21.link/p0214r8>.
- [P0964R0] Matthias Kretz. *P0964R0: Finding the right set of traits for `simd<T>`*. ISO/IEC C++ Standards Committee Paper. 2018. URL: <https://wg21.link/p0964r0>.
- [P0820R1] Tim Shen. *P0820R1: Feedback on P0214R5*. ISO/IEC C++ Standards Committee Paper. 2017. URL: <https://wg21.link/p0820r1>.